

JetpackCompose-WeatherApp Documentation

1. Introduction

JetpackCompose-WeatherApp is an Android application developed using **Kotlin** and **Jetpack Compose**. It provides real-time weather updates for the user's current location and allows searching for weather conditions in other cities. The app is designed with **modern UI principles**, supports **offline caching**, and automatically updates weather data when an internet connection is available.

2. Features & Functionalities

2.1 Basic Functionalities

The application meets the following **basic requirements**:

✓ Multiple Activities / Fragments

- The app follows the MVVM architecture pattern with **multiple Composable functions** acting as fragments.
- Navigation is implemented using **Jetpack Navigation Component**.

✓ Network & Database Connectivity

- Fetches real-time weather data from **OpenWeatherMap API**.
- Uses **Retrofit** for network calls.

✓ Caching & Background Services

- Stores weather data locally using **Room Database**.
- Uses **WorkManager** for periodic background updates.

✓ Error Handling

- Displays appropriate **error messages** when the network is unavailable.
 - Implements a **retry mechanism** for failed API calls.
-

2.2 Extra Features

The app incorporates the following **additional features**:

- ◆ **Sensing (Native API Usage)**
 - Utilizes **Fused Location Provider API** to detect the user's current location for weather updates.
 - ◆ **Unique User Interface (Jetpack Compose)**
 - Built entirely using **Jetpack Compose**, ensuring a **modern, declarative UI**.
 - **Dynamic theming** (light & dark mode) based on system settings.
 - **Animations & transitions** for a smooth user experience.
-

3. Technologies Used

3.1 Programming Languages & Frameworks

- **Kotlin** (Primary Language)
- **Jetpack Compose** (UI Framework)
- **MVVM Architecture** (ViewModel, LiveData, Repository Pattern)

3.2 Libraries & APIs

Library/API	Purpose
Retrofit	HTTP client for API calls
Fused Location Provider API	Fetch user location
Room Database	Local storage for weather data
Jetpack Navigation Component	Manages in-app navigation
WorkManager	Background data fetching
Coil	Image loading
Coroutines & Flow	Asynchronous programming

4. System Requirements

- **Minimum Android Version:** Android 6.0 (API Level 23)
 - **Target Android Version:** Android 14 (API Level 34)
 - **Required Permissions:**
 - `ACCESS_FINE_LOCATION` (For location-based weather updates)
 - `INTERNET` (For fetching weather data)
-

5. Installation & Setup

5.1 Cloning the Repository

```
git clone https://github.com/username/JetpackCompose-WeatherApp.git
cd JetpackCompose-WeatherApp
```

5.2 Setting Up API Key

1. Register on [OpenWeatherMap](#) and get an API key.

Add the API key to `local.properties`:

```
WEATHER_API_KEY=your_api_key_here
```

- 2.

5.3 Running the App

1. Open the project in **Android Studio**.
 2. Sync Gradle files.
 3. Run the app on an **emulator or physical device**.
-

6. Application Flow

1. **Splash Screen** → Displays the app logo.
 2. **Home Screen** → Shows current weather based on location.
 3. **Search Screen** → Allows users to find weather for other cities.
 4. **Settings Screen** → Manage preferences (e.g., dark/light mode).
-

7. Future Enhancements

- ✓ Add **ML-based weather predictions** using TensorFlow Lite.
 - ✓ Improve **energy efficiency** by optimizing background updates.
 - ✓ Implement **voice search functionality** for city lookup.
-

8. Conclusion

JetpackCompose-WeatherApp is a **modern, feature-rich weather application** that utilizes Android's **native APIs**, **sensing capabilities**, and **local caching** to deliver an efficient and seamless user experience. With **Jetpack Compose** powering the UI, the app ensures a smooth, visually appealing, and responsive design.

For more details, visit the **GitHub Repository**: [JetpackCompose-WeatherApp](#).